

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

Phase 2: Planning phase

[SMART FUNERAL CLAIMS PLATFORM]

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Introduction

The planning phase is an important stage in the development of the Smart Funeral Claims Platform. During this phase, the project is planned in detail before the main development and implementation takes place. The purpose of this phase is to understand the problem, identify the needs of the users, determine the resources required, and make sure that the project is possible to complete within the available time and resources.

The Smart Funeral Claims Platform is a web-based system designed to make the funeral insurance claims process easier and more efficient. The system will allow users to register, log in, submit funeral claims, upload supporting documents and track their claim status. An administrator will also be able to manage users and review and update claims through an admin dashboard.

The planning phase includes the identification of the need for the system, preliminary investigation, feasibility study, project planning, project scheduling using PERT and Gantt charts, Software Requirements Specification (SRS), and data modelling. These activities provide a clear plan for the development of the system. The planning phase therefore creates a foundation for the next stages of the project. This follows the structure used in the provided planning-phase guideline.

2.1 Identification of need

The funeral insurance claims process can be difficult when it depends heavily on paper forms, physical documents and manual communication. Users may need to visit an office to submit documents, while administrators have to manually check and process the information. This can result in delays, lost documents, mistakes and difficulty tracking the progress of a claim.

The Smart Funeral Claims Platform provides an opportunity to improve this process by moving the claims process to an online platform.

Main purpose

The main purpose of the project is to develop a system that allows users to:

1. Create an account (Misure, 2023).
2. Log into the system.
3. Submit funeral insurance claims online.
4. Upload supporting documents.
5. View the status of their claims.
6. Communicate with the organization through the contact form.

(Misure, 2023).

The system will also provide an administrator dashboard where administrators can:

1. View registered users.
2. View submitted claims.
3. Review claim information.
4. View supporting documents.
5. Update claim statuses.
6. Manage the claims process.

(Misure, 2023).

The project therefore aims to make the claims process faster, more organized and easier for both users and administrators (Misure, 2023).

Who is it designed for?

The main beneficiaries of the Smart Funeral Claims Platform are:

Policyholders and Claimants

Users will be able to submit claims without having to rely completely on physical paperwork. They will also be able to check their claim status online (Misure, 2023).

Administrators

Administrators will have a central system where they can view and manage claims and users. This can reduce the amount of manual paperwork they have to handle (Misure, 2023).

Funeral Insurance Organizations

The organization can benefit from a more organized claims process, better document management and improved communication with customers (Misure, 2023).

Families

Families dealing with funeral arrangements may benefit from a simpler claims process during a difficult time (Misure, 2023).

Why is it a need?

The project is necessary because the funeral claims process can involve sensitive information and important supporting documents. If the process is mostly manual, it can take longer to process claims and make it difficult for users to know what is happening with their claim (MiWay, 2024). The proposed system provides a central online platform where claims and documents can be submitted and managed. It also gives users better visibility of their claim progress (MiWay, 2024). The system is especially useful because it combines user registration, authentication, claim submission, document uploading, claim tracking and administrator management into one platform (MiWay, 2024).

Reason

The Smart Funeral Claims Platform is useful because it addresses problems associated with manual funeral insurance claims. The system can reduce paperwork, make information easier to manage and improve communication between users and administrators.

2.2 Preliminary investigation

Project Goal

The main goal of the project is to design and develop a web-based platform that simplifies the funeral insurance claims process.

The system aims to:

1. Provide secure user registration and login.
2. Allow users to submit funeral claims online.
3. Allow users to upload supporting documents.
4. Allow users to track claim progress.
5. Provide an administrator dashboard.

6. Allow administrators to manage claims and users.
7. Provide a simple and user-friendly interface.

(Inabo, 2025).

Project Scope

The system will focus on the digital submission and management of funeral insurance claims.

The system will include:

- User registration.
- User login.
- User dashboard.
- Claim submission.
- Document uploads.
- Claim tracking.
- Administrator login.
- Administrator dashboard.
- Claim management.
- User management.
- Contact form.

(Inabo, 2025).

The system will not directly make insurance payout decisions. The administrator or relevant organization will still be responsible for the final decision regarding a claim (Inabo, 2025).

Project constraints

Some possible challenges include:

- Firebase configuration problems.
- Authentication errors.
- Problems connecting the frontend to the backend.
- File upload problems.
- Internet connection problems.
- Protecting sensitive user information.
- Making sure the system works correctly on different browsers and screen sizes.

(Inabo, 2025).

What is required for the project?

The initial requirements for the system were identified by studying the funeral claims process and considering the problems that users and administrators may experience (Inabo, 2025).

The main requirements identified are:

- Users need to create accounts.
- Users need to log into the system.
- Users need to submit claims electronically.
- Users need to upload supporting documents.
- Users need to track claims.
- Administrators need to view claims.
- Administrators need to update claim statuses.
- Administrators need to manage users.
- The system needs to protect user information.

(Inabo, 2025).

These requirements will guide the design and development of the system.

Project partakers

Users/Claimants – These are the people who will use the system to register, submit claims, upload documents and track their claims (Inabo, 2025).

Administrators – Administrators will use the system to manage users, view claims, review information and update claim statuses (Inabo, 2025).

Insurance Organization – The organization is responsible for the funeral insurance services and will benefit from having a more organized claims process.

Project Developer – Including myself as the developer, my responsibility is to is responsible for analyze, design, develop, test and implement the system (Inabo, 2025).

The Smart Funeral Claims Platform will require different resources to successfully develop and implement the system. The main hardware resource needed is a laptop or desktop computer with enough processing power and storage to run Visual Studio Code, Node.js, Firebase tools and a web browser at the same time (Aimen, 2026). A stable internet connection will also be needed because the system uses Firebase services for authentication, database storage and document storage. A smartphone can also be used during testing to check whether the website displays correctly on different screen sizes (Johns, 2025).

The main software resources required for the project include Visual Studio Code, which will be used to write and manage the source code, and Node.js, which will be used to run the backend of the system. The project will also use HTML, CSS and JavaScript for the frontend, while Node.js and Express.js will be used for the backend (Aimen, 2026). Firebase will be used for authentication, Firestore database services and file storage. Git and GitHub can also be used to manage the source code and provide a backup of the project. A web browser such as Google Chrome or Microsoft Edge will be required to run and test the system (Johns, 2025).

The project will also require human resources. The developer will be responsible for analyzing the problem, gathering requirements, designing the system, developing the frontend and backend, connecting Firebase services, testing the system and correcting errors. Users and administrators will also be considered during the development process because their needs will help determine the functions and design of the system (Patel, 2026).

Overall, the resources required for the Smart Funeral Claims Platform are available, most of the software technologies used are free or have free development options, which makes it possible to develop and test the system without requiring expensive equipment or software.

The development of the Smart Funeral Claims Platform may have some risks and constraints that could affect the project. One of the main risks is technical problems when connecting the frontend, backend and Firebase services (Patel, 2026). For example, incorrect Firebase configuration could cause problems with user registration, login, database access or document uploads. Another risk is that the system depends on an internet connection because Firebase is a cloud-based service (Johns, 2025). If the internet connection is unavailable, some parts of the system may not work correctly. Security is also an important risk because the system will handle user information, claim details and supporting documents. Unauthorized access to this information could cause privacy problems. Another possible risk is limited development time, which may make it difficult to complete and test every planned feature. To reduce these risks, the system will be tested regularly, important configuration settings will be checked carefully, and the project will be divided into smaller tasks that can be completed within the available time (Patel, 2026).

2.3 Feasibility study

Technical feasibility

The Smart Funeral Claims Platform is technically feasible because the technologies required to develop the system are available and suitable for the project. The frontend will be developed using HTML, CSS and JavaScript, while Node.js and Express.js will be used for the backend. Firebase will be used for services such as user authentication, database management and storage of uploaded documents (Martins, 2026).

Visual Studio Code will be used as the main development environment. These technologies are commonly used for web application development and can work together to provide the required functions of the system. The developer will also have access to the required computer, internet connection and development software. Therefore, the project can be developed using the available technical resources (Martins, 2026).

Operational feasibility

The Smart Funeral Claims Platform is operationally feasible because the proposed system is designed to be simple and easy to use. Users will be able to access the system through a web browser and will not need specialized software. A user will be able to create an account, log in, submit a claim, upload supporting documents and check the status of the claim through the website (Martins, 2026).

Administrators will use the admin dashboard to manage users and claims. The system is expected to reduce the amount of manual work involved in the claims process and make information easier to access and manage. Because the system is designed around the activities normally performed by users and administrators, it can be integrated into the claims process with limited training (Martins, 2026).

Economic feasibility

The Smart Funeral Claims Platform is economically feasible because the project can be developed using mostly free or low-cost development technologies. Visual Studio Code, Node.js, HTML, CSS and JavaScript can be used without purchasing expensive development software. Firebase also provides services that can be used during development and testing without requiring a large initial investment (Aimen, 2026).

Git and GitHub can be used to manage and back up the source code. The main costs may come from internet access and any future Firebase services required if the system grows and exceeds the available free usage. For the purpose of the student project, the expected development costs are therefore relatively low (Aimen, 2026).

2.4 Project planning

Week	Phase	Tasks Completed
Week 1	Project Initiation	The project idea was identified and the main problem with the funeral insurance claims process was investigated. The purpose and main objectives of the Smart Funeral Claims Platform were established.
Week 2	Requirements Gathering	The main users of the system were identified, including claimants and administrators. Requirements such as registration, login, claim submission, document uploading, claim tracking and administration were identified.
Week 3	System Planning	The project scope and main features were defined. The resources, technologies, possible risks and limitations of the project were considered.
Week 4	System Design	The structure of the system was planned. The main pages, user interface, database requirements, system architecture and administrator dashboard were considered.

Week 5 - 7	Frontend Development	Development of the website interface started using HTML, CSS and JavaScript. Pages such as the home page, registration page, login page and other user pages were created and styled.
Week 8 - 10	Backend Development	The Node.js and Express backend was developed. The backend was configured to run locally, and the required packages and dependencies were installed and configured.
Week 11	Firebase Integration	Firebase services were configured for the project. Firebase Authentication, Firestore and Firebase Storage were connected to support user accounts, data storage and document uploads.
Week 12	User Authentication	The registration and login functions were developed and tested. Users were given the ability to create accounts and access the system using their login details. Authentication problems were identified and worked on during testing.
Week 13	Claims Development	The claim submission functionality was developed. The system was designed to allow users to enter claim information and submit their claims electronically.
Week 14	Document Upload and Tracking	Document uploading and claim tracking features were developed. Users were provided with functionality to upload supporting documents and view information about their submitted claims.
Week 15	Admin Dashboard	The administrator section of the system was developed. The admin dashboard was designed to

		allow administrators to view users and claims and manage claim information.
Week 16	System Integration	The different parts of the system were connected together. The frontend, Node.js backend and Firebase services were tested together to make sure information could move correctly between the different components.
Week 17	Testing and Debugging	The system was tested to identify problems with registration, login, claim submission, document uploading and administrator functions. Errors were investigated and corrections were made where possible.
Ongoing	Documentation and Presentation Preparation	Project documentation was prepared, including the planning, analysis and system development information. The system architecture, project features and development process were reviewed in preparation for the final presentation.

Table 1: Project planning table

2.5 Project Scheduling

PERT Chart

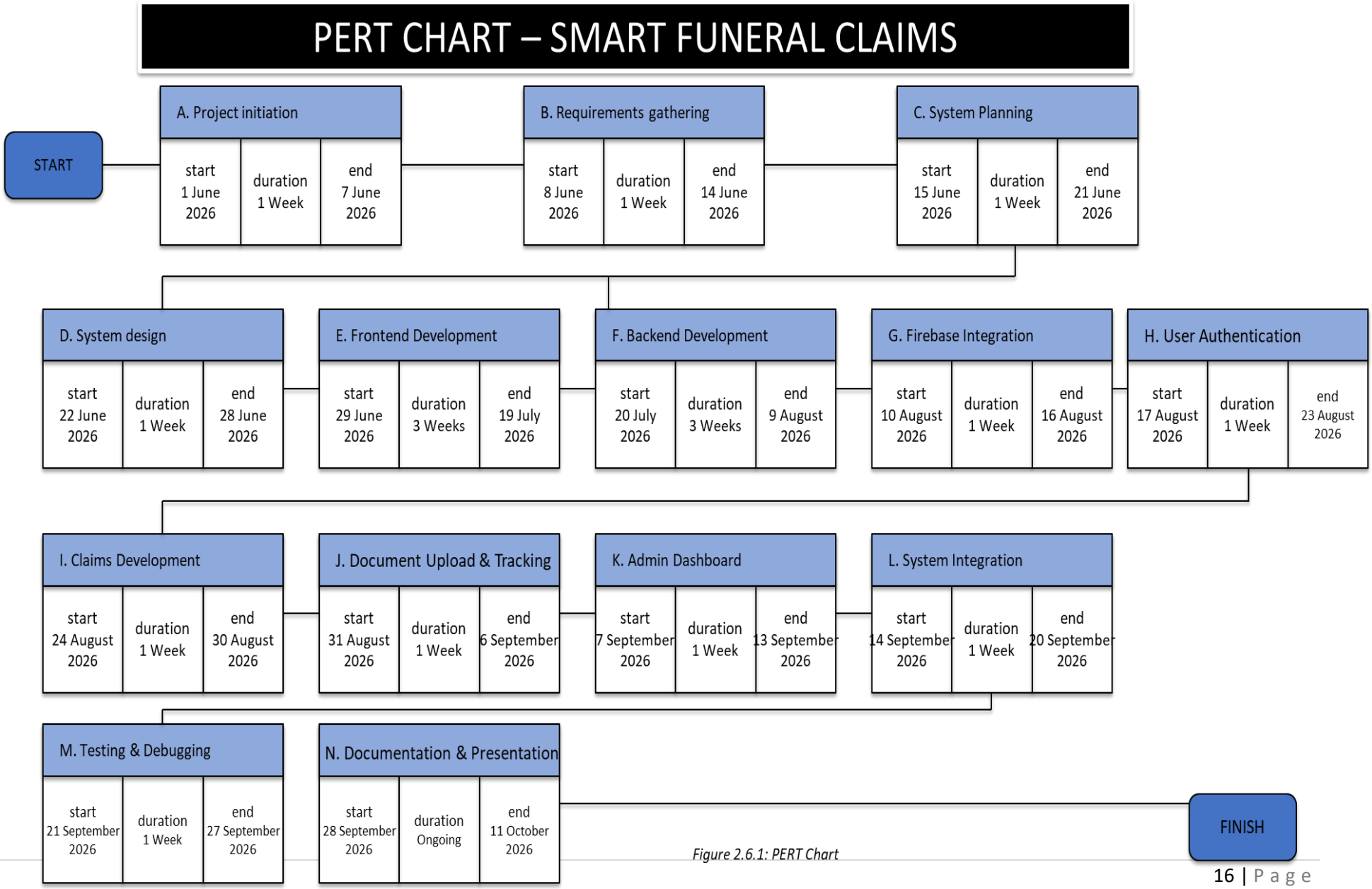


Figure 2.6.1: PERT Chart

Gantt Chart

GANTT CHART – SMART FUNERAL CLAIMS PLATFORM

Phase	Task	Start Week	Duration	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17
Timeline	*June–October 2026*			01 Jun	08 Jun	15 Jun	22 Jun	29 Jun	06 Jul	13 Jul	20 Jul	27 Jul	03 Aug	10 Aug	17 Aug	24 Aug	31 Aug	07 Sep	14 Sep	21 Sep
Planning	Project Initiation	1	1	■																
Planning	Requirements Gathering	2	1		■															
Planning	System Planning	3	1			■														
Planning	System Design	4	1				■													
Development	Frontend Development	5	3					■	■	■										
Development	Backend Development	8	3								■	■	■							
Development	Firebase Integration	11	1											■						
Development	User Authentication	12	1												■					
Development	Claims Development	13	1													■				
Development	Document Upload & Tracking	14	1														■			
Development	Admin Dashboard	15	1															■		

Integration & Testing	System Integration	16	1															■	
Integration & Testing	Testing & Debugging	17	1																■
Finalization	**Documentation & Presentation Preparation**	1	17	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

Legend: ■ **Planned activity** ■ **Ongoing documentation**

Project scheduling

Phase	Task	Start Week	Duration	End Week
Planning	Project Initiation	1	1	1
Planning	Requirements Gathering	2	1	2
Planning	System Planning	3	1	3
Planning	System Design	4	1	4
Development	Frontend Development	5	3	7
Development	Backend Development	8	3	10
Development	Firebase Integration	11	1	11
Development	User Authentication	12	1	12
Development	Claims Development	13	1	13
Development	Document Upload & Tracking	14	1	14
Development	Admin Dashboard	15	1	15
Integration & Testing	System Integration	16	1	16
Integration & Testing	Testing & Debugging	17	1	17
Finalization	Documentation & Presentation Preparation	1	17	17

2.6 Software requirements specification

Functional Requirements

User Registration

1. Allow new users to create an account.
2. Validate the information entered by the user.
3. Store user authentication information securely.

(Sana, 2026).

Login

4. Allow registered users to log into the system.
5. Prevent unauthorized users from accessing protected pages.
6. Allow users to log out.

(Sana, 2026).

Claims

7. Allow users to submit funeral insurance claims online.
8. Allow users to enter the required claim information.
9. Allow users to upload supporting documents.

10. Store claim information in the database.
11. Allow users to view the status of their claims.

(Sana, 2026).

Administrator

12. Allow administrators to log into the admin dashboard.
13. Allow administrators to view submitted claims.
14. Allow administrators to review claim information.
15. Allow administrators to update claim status.
16. Allow administrators to manage users.

(Sana, 2026).

Contact

17. Allow users to send enquiries through the contact form.
18. Allow the system to send contact information through the configured email service.

(Sana, 2026).

Non-Functional requirements

Security

The system should protect user information and prevent unauthorized access.

Usability

The system should have a simple and easy-to-understand interface.

Performance

The website should respond quickly when users perform normal operations.

Reliability

The system should save information correctly and avoid unnecessary data loss.

Availability

The system should be available to users whenever the required online services are working.

Maintainability

The code should be organized so that future changes and improvements can be made easily.

Compatibility

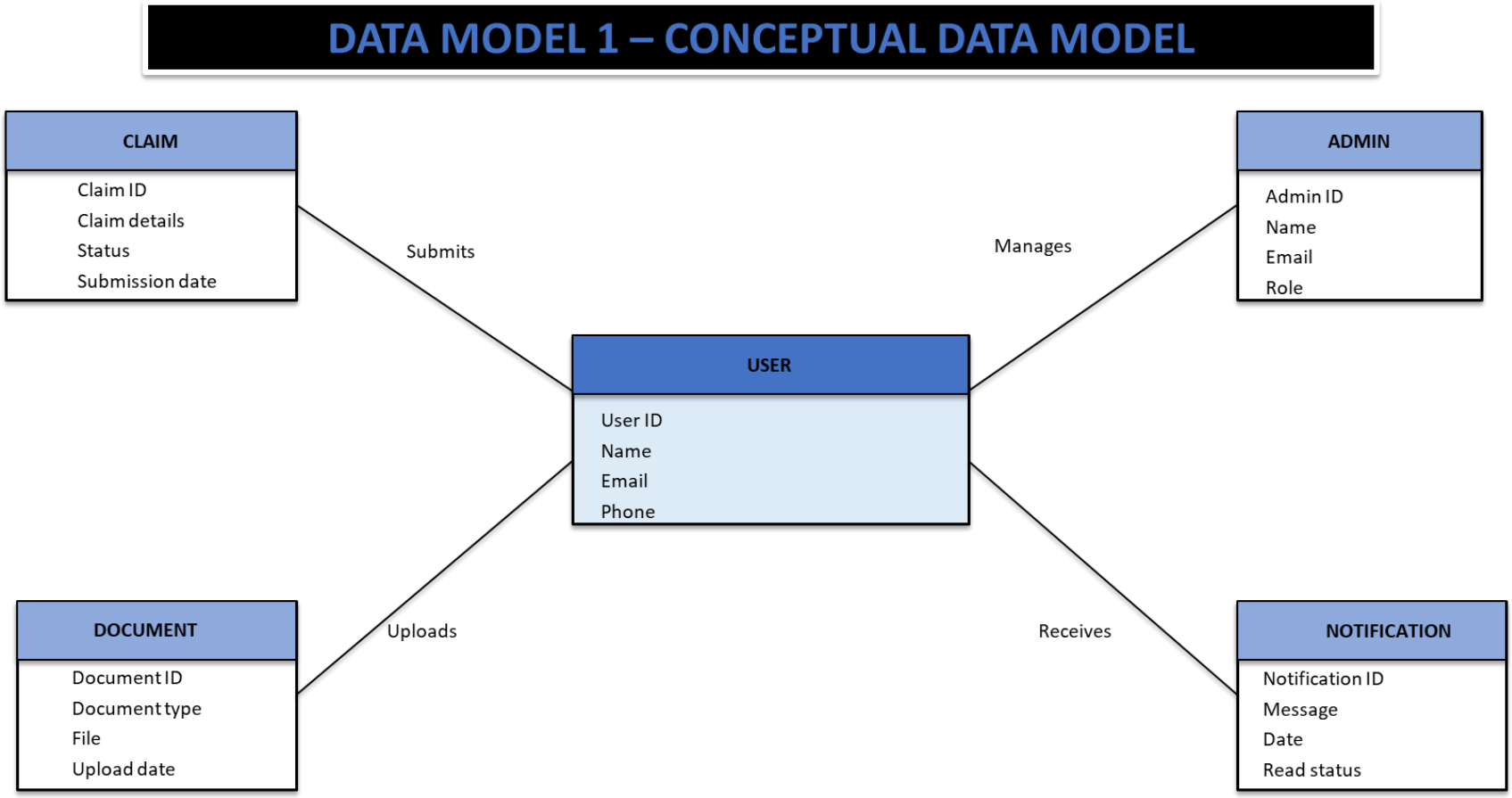
The website should work correctly on common modern web browsers such as Chrome, Edge and Firefox.

Scalability

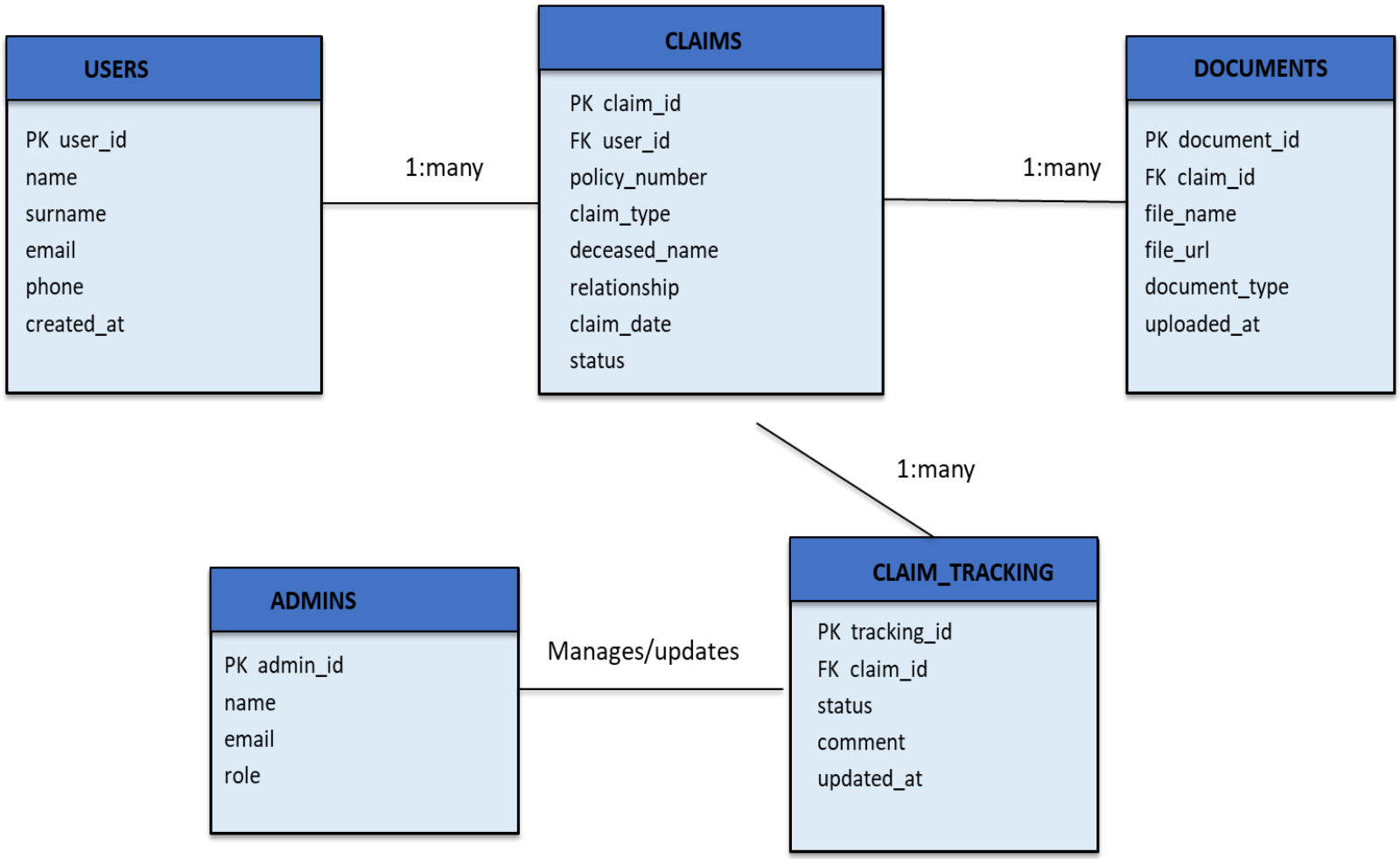
The system should be able to support more users and claims as the project grows.

(Sana, 2026).

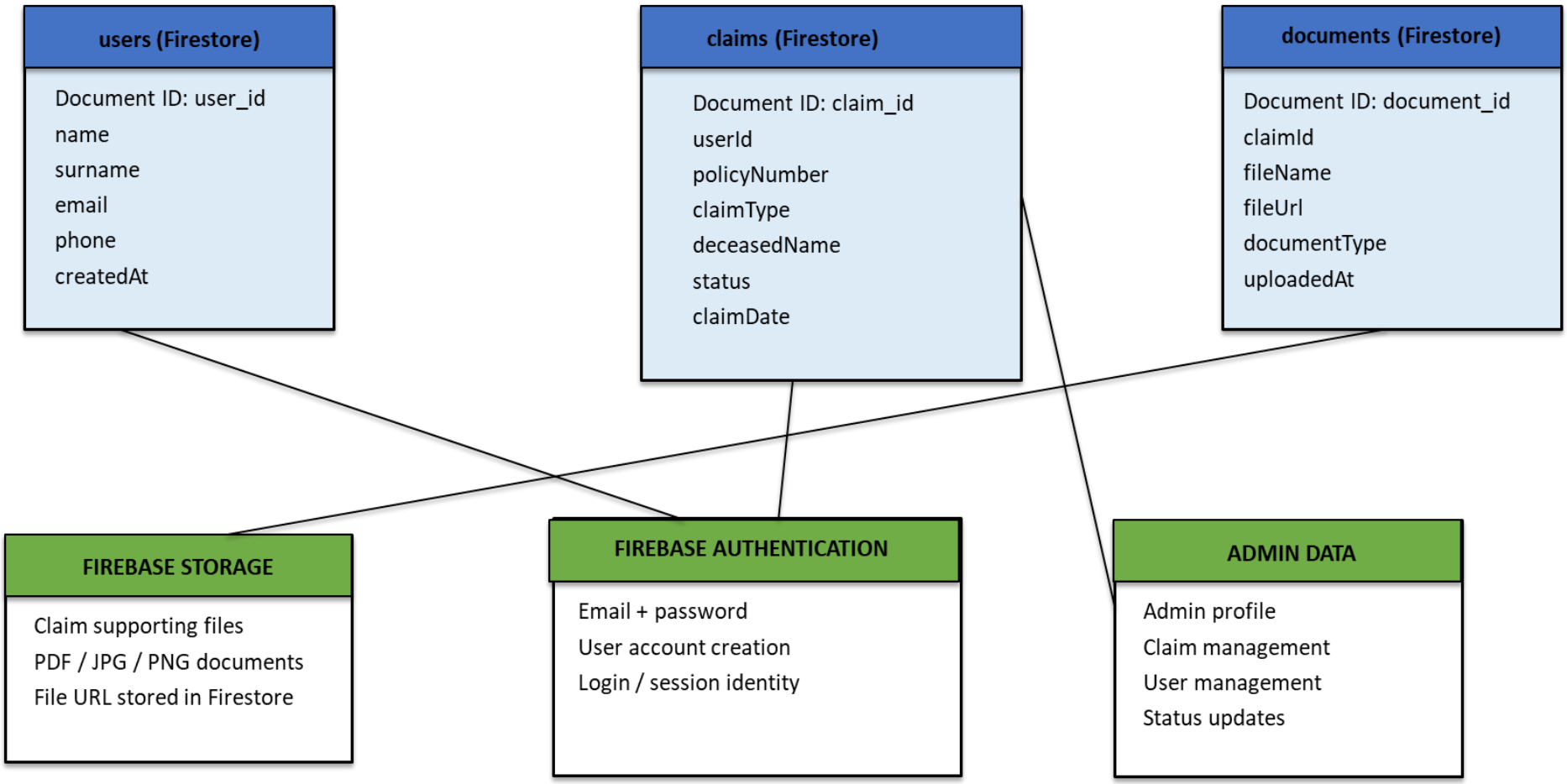
2.7 Data Models



DATA MODEL 2 – LOGICAL DATA MODEL



DATA MODEL 3 – FIREBASE / PHYSICAL DATA MODEL



DATA MODEL 4 – CORE DATA DICTIONARY

Entity	Field	Type	Description
Users	user_id	String	Unique user identifier
Users	name	String	User first name
Users	surname	String	User surname
Users	email	String	User email address
Claims	claim_id	String	Unique claim identifier
Claims	user_id	String	Links claim to the claimant
Claims	policy_number	String	Funeral insurance policy number
Claims	claim_type	String	Type/category of claim
Claims	deceased_name	String	Name of deceased person
Claims	status	String	Current claim status
Documents	document_id	String	Unique document identifier
Documents	claim_id	String	Links document to a claim
Documents	file_name	String	Uploaded document filename
Documents	file_url	String	Firebase Storage file location
Documents	document_type	String	Type of supporting document
Admins	admin_id	String	Unique administrator identifier
Admins	role	String	Administrator access role

Conclusion

The planning stage has laid out a detailed strategy for creating the Smart Funeral Claims Platform. The necessity for this system was recognized due to the issues associated with manual funeral insurance claims, including paperwork complications, delays, and challenges in tracking claims. The initial investigation pinpointed the primary users, necessary resources, project scope, risks, and potential challenges. The feasibility study further confirmed that the project is viable from technical, operational, and economic standpoints, utilizing technologies such as HTML, CSS, JavaScript, Node. js, Express. js, and Firebase.

The project timeline outlines a clear schedule for completing various tasks, while the PERT and Gantt charts will assist in tracking the project's advancement. The Software Requirements Specification details both functional and non-functional requirements, whereas the data model illustrates the relationships among the main entities within the system. In summary, the planning phase establishes a solid groundwork for transitioning to the next step of the project, namely the Analysis and Design Phase. This aligns with the objectives of the planning phase as outlined in the provided guidelines, which focus on defining requirements, scope, resources, timelines, feasibility, and data structure prior to development.

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b842a9eb6528&psq=steps+for+project+planning&u=a1aHR0cHM6Ly93d3cuZmxvYXQuY29tL3Jlc291cmNlcy9wcm9qZWN0LXBsYW5uaW5nLXByb2Nlc3M [Accessed 12 August 2026]

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